

TokenEconomics Studio user guide

Connected RAG walkthrough | Azure-hosted edition | September 14, 2026

This guide follows **Gutenberg books RAG - live Foundry playground**, report `RPT-20260908-CA037FE9`, through the current Studio. It explains each screen, the evidence it displays, and why that evidence matters.

All screenshots are fresh captures of the deployed application on September 14, 2026. The example uses previously saved forecasts and the completed September 9 batch. No forecasts were generated, questions executed, policies changed, billing synced, feedback registered, or advisory reviews saved to produce this guide. Browser navigation and disclosure panels were used with non-read requests blocked.

Open [TokenEconomics Studio on Azure](#). The [PDF edition](#) contains the same walkthrough.

Read the labels before interpreting the numbers. A forecast is modeled. Completed responses are measured usage, not accepted answers. A catalog-priced model allocation is not an invoice. An imported resource bill is not automatically attributable to a batch. A saved review is advisory, not permission to execute.

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1. Home: open the existing report

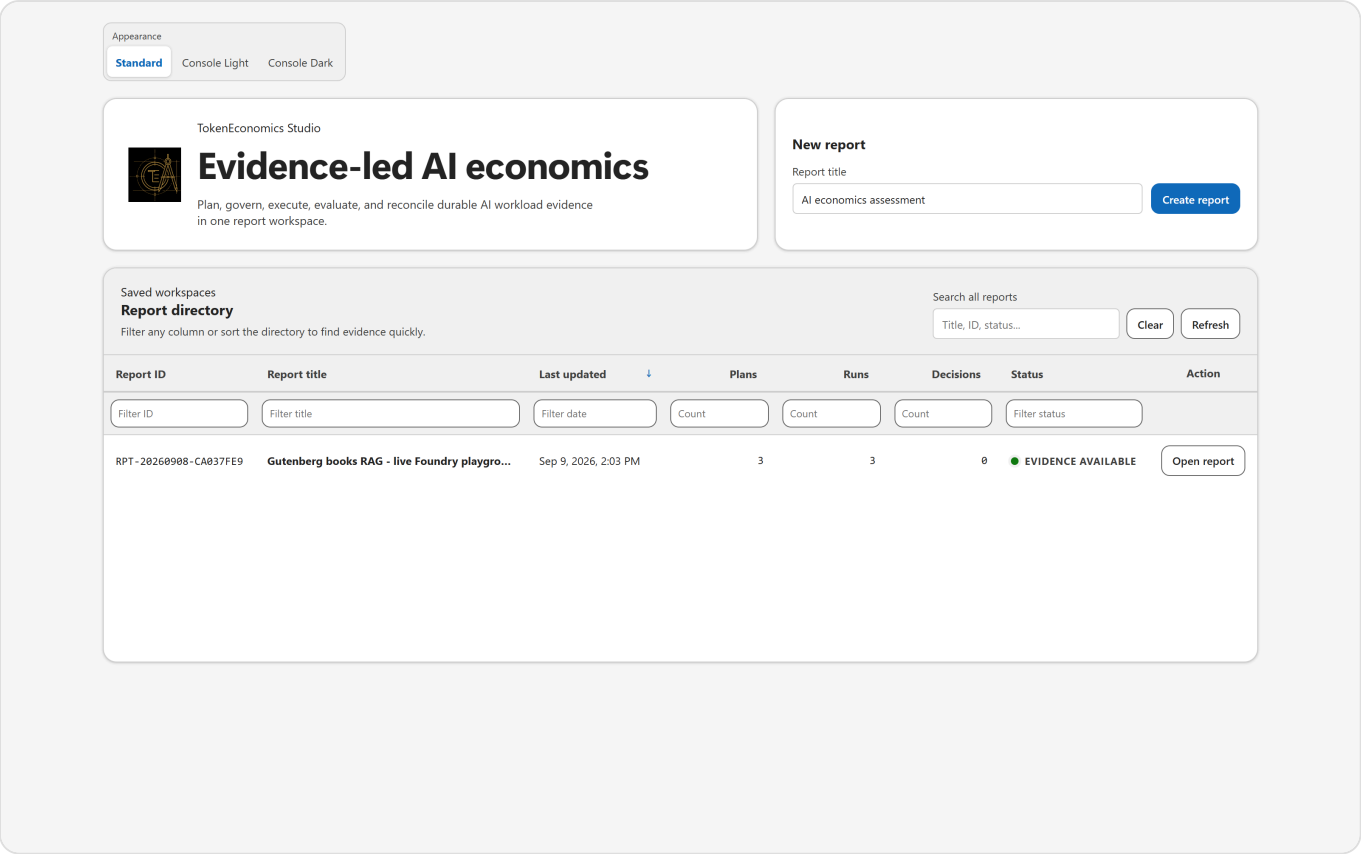


Figure 1. The current report directory, with the connected RAG report and its three saved forecasts and three runs.

Select **Open report** on `RPT-20260908-CA037FE9` . Opening a report restores its latest saved forecast; it does not repeat estimation or call an AI model.

Feature	What it shows or does	Why it matters
Report directory	Report identity, title, last update, artifact counts and status.	Keeps one investigation together rather than comparing unrelated runs.
Search, column filters and sortable headings	Find or order saved workspaces; Clear removes filters.	Useful as the directory grows. None creates new evidence.
Plans / Runs / Decisions	Counts in the directory's corresponding record categories.	These are inventory counts, not acceptance scores. The Decisions column is not a total of every advisory or billing-feedback record.
Evidence available	The report has saved evidence to inspect.	Does not mean quality was accepted, billing is complete, or execution is currently authorized.
New report / Create report	Creates another durable workspace.	A state-changing action; not used in this walkthrough.
Standard / Console Light / Console Dark	Browser appearance choices.	Changes presentation, not receipts, prices or policy.

Home intentionally does **not** show Project history. Cross-report experiments are at the bottom of **Performance & Decisions**.

2. Shared context and navigation

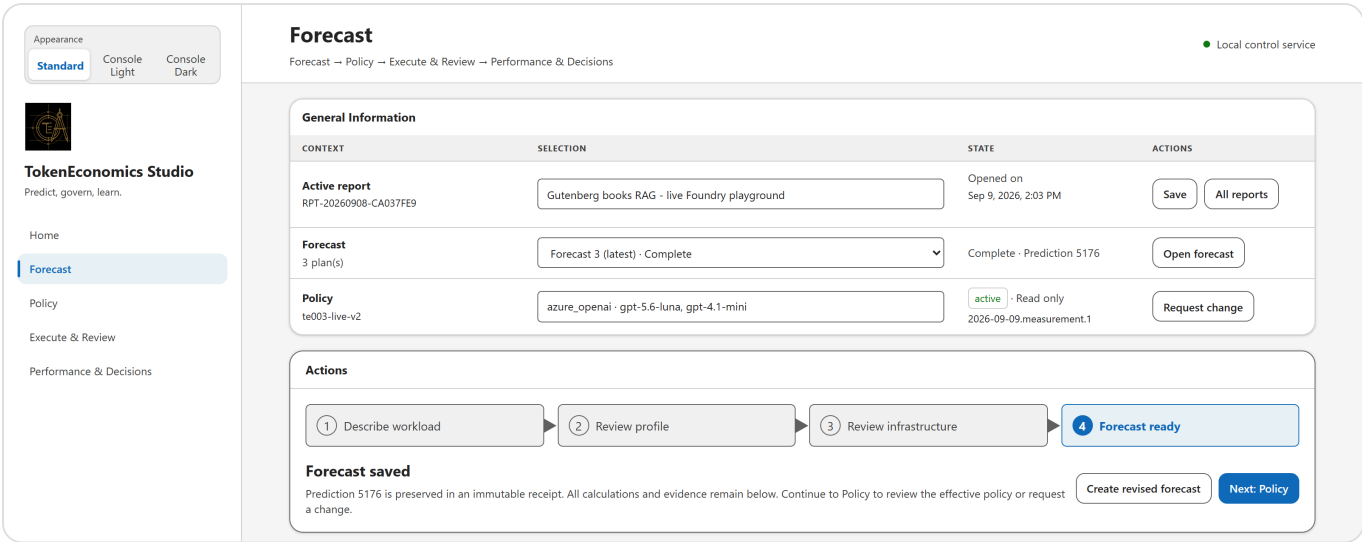


Figure 2. Opening the report selects Forecast 3 and preserves prediction 5176.

The sidebar follows **Forecast -> Policy -> Execute & Review -> Performance & Decisions**. Home is outside that workflow. Selecting a tab navigates; it does not authorize or start the associated business operation.

Every workflow page begins with **General Information**, followed by **Actions**:

Area	How to read it
Active report	The report ID is stable; the title is descriptive. Save changes report metadata and is not needed for inspection.
Associated forecasts	Three revisions are available. Use Forecast 3 (latest) , prediction 5176 , for this example. Open forecast restores a selected revision; it does not overwrite another.
Policy	Shows the current Azure policy label, allowed supply and version. Read the measurement expiry on the Policy page separately.
Actions	Groups the current page's navigation and explicit operations. A visible action does not imply all prerequisites are satisfied.
Local control service	Legacy connection wording. In these screenshots, the connected Studio service runs in Azure Container Apps, not on the reader's computer.

The four connected Forecast stages are **Describe workload**, **Review profile**, **Review infrastructure**, and **Forecast ready**. The last stage is selected because this forecast is already saved. **Forecast saved** means its immutable receipt exists, not that quality or execution has been approved.

Saved forecast inputs remain read-only. **Create revised forecast** begins a new revision workflow; it was not selected here. Earlier forecasts and their original model choices remain historical evidence.

Dates in these screenshots use **US Eastern time**. Report dates, batch start dates, source delivery dates and authorization expiry are different timestamps; the stored UTC evidence remains authoritative.

3. Forecast: delivery, model and workload profile

Scroll below Actions to inspect the saved description and delivery choice. The description identifies a deployed Foundry agent answering questions about five Project Gutenberg books using an Azure AI Search-backed knowledge source.

How will this work be delivered?

Microsoft Copilot for employees

Chat and in-app assistance grounded in work data.

subscription included

Planned for a future release

Copilot Cowork

Delegate complex, long-running work to Cowork.

subscription native credit

Planned for a future release

Agent Builder

Create a lightweight Microsoft 365 knowledge agent.

subscription included

Planned for a future release

Copilot Studio

Build and operate a custom agent or workflow.

native credit included

Planned for a future release

Custom agent using Work IQ APIs

Add Microsoft 365 context and actions to another agent.

native credit

Planned for a future release

Microsoft Foundry

Build a pro-code custom AI application or agent.

direct token resource

GitHub Copilot

Forecast token-derived GitHub AI Credits for developer use.

subscription token derived credit resource

Planned for a future release

Copilot Studio with a Foundry model

Compose Copilot Studio credits with a separate model forecast.

native credit direct token

Planned for a future release

Microsoft Foundry with Work IQ APIs

Compose model usage with separate Work IQ API credits.

direct token native credit

Planned for a future release

Figure 3. The hosted release exposes Microsoft Foundry; the other eight options remain visible with future-release help.

Microsoft Foundry is the only available delivery choice in the hosted release. The saved form itself is read-only. The eight future routes are Microsoft Copilot for employees, Copilot Cowork, Agent Builder, Copilot Studio, custom agents using Work IQ APIs, GitHub Copilot, Copilot Studio with a Foundry model, and Microsoft Foundry with Work IQ APIs.

Their visibility is not a claim of eight working hosted integrations. Native local Studio retains broader modeled route support; this guide documents Azure-hosted behavior. Microsoft Copilot Credits, GitHub AI Credits, subscriptions, direct model tokens and resource units are distinct meters, not interchangeable balances.

Default / coordinator model
Users
Calls per user / day

GPT-4.1 Mini
10
3

Azure region

eastus

OpenAI on Azure
Input \$0.40 / 1M
Output \$1.60 / 1M
Global Standard
Release 2026-08-25.2
1,048,576 context
text · tools

[Exact pricing evidence](#)
Govern eligible under policy 2026-09-09.measurement.1 · te003-live-v2

▶ All supported OpenAI and Anthropic Foundry models

Review inferred workload profile
Modeled intake

Analysis 1.0 · rules enterprise-semantics-2026-09-08.1 · confidence medium

Topology
Agent count
Modalities, comma separated

rag pipeline
1
text, document

Tools, comma separated

file_search

Alternatives: None · **Agent-count source:** defaulted

▶ Why this profile was inferred

Resolve before estimating:

- Confirm defaulted cost-material quantity assumptions before estimation.

Reviewing these fields confirms modeled workload-invocation assumptions. It does not validate runtime behavior, quality, complete trajectory cost, or tail-risk calibration.

▶ Use a different model for each agent

Figure 4. Saved model and workload assumptions, shown without editing them.

Field or panel	Value in this forecast	Meaning and importance
Default / coordinator model	GPT-4.1 Mini	The model offering used to price this forecast; not merely the cheapest current catalog entry.
Users / calls per user per day	10 / 3	Forecast scale assumptions, not measured traffic.
Azure region	eastus	Planning/pricing region. It is not a live resource inventory; the Studio host itself is in eastus2.
Model detail strip	Input \$0.40 and output \$1.60 per million tokens; Global Standard	Exact catalog pricing and offering context. A catalog release date is different from the deployed model version.
Topology	rag pipeline	Selects the applicable workload experience from the saved forecast.
Agent count / modalities	1 / text, document	Modeled structure and token categories, not proof of internal agent steps.

Field or panel	Value in this forecast	Meaning and importance
Tools	file_search	A forecast assumption. Generic file-search pricing is not automatically valid for the deployed managed knowledge-base tool.
Why this profile was inferred	Versioned analysis and rules	Makes defaults and inference explainable instead of silently treating prose as fact.
Optional per-agent models	Not used for this one-agent example	Allows modeled allocations in suitable multi-agent forecasts; turn weights are not measured runtime turns.

Why this screen matters: forecasts are only meaningful with their assumptions. Checking topology, retrieval context, scale and tool applicability is more useful than reading an isolated dollar figure.

4. Forecast: read the saved economics

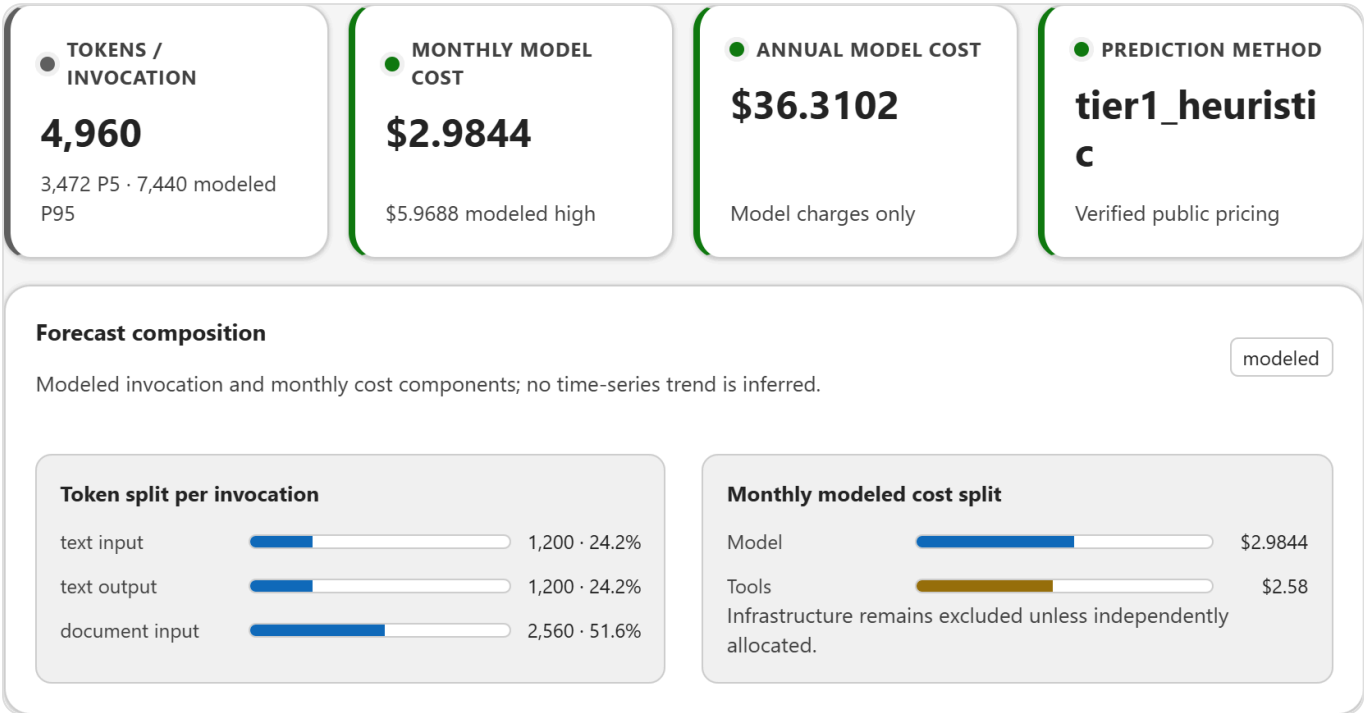


Figure 5. The forecast separates token demand, model costs and generic tool costs.

Saved result	Value	Interpretation
Tokens per invocation	4,960	Modeled total, not an observed complete agent trajectory.
Text input / document input / text output	1,200 / 2,560 / 1,200	The document context assumption contributes to modeled model input.
P5 / P50 / P95 tokens	3,472 / 4,960 / 7,440	Heuristic distribution points, not calibrated budget-breach guarantees.
Monthly model mean	USD 2.9844	Model-only estimate under the saved scale and cache assumptions.
Annual model mean	USD 36.3102	Uses 365 days, whereas monthly uses 30 days; do not assume annual equals monthly times 12.
Generic tool estimate	USD 2.58/month	Separate from model costs and subject to tool-scope applicability.
Prediction method	tier1_heuristic	This saved prediction is not a calibrated response-usage forecast.

The composition charts show the token mix and separate monthly modeled components. They are not historical time-series charts. Infrastructure is excluded from those model/tool bars and has its own section.

▼ Model cost forecast evidence				
Period	Low	Mean	High	Modeled high
Per invocation	\$0.0024	\$0.0034	\$0.0051	\$0.0068
Daily	\$0.0696	\$0.0995	\$0.1492	\$0.199
Monthly	\$2.0891	\$2.9844	\$4.4766	\$5.9688
Annual	\$25.4171	\$36.3102	\$54.4653	\$72.6204
► Show calculation				

Figure 6. Low, mean, high and modeled-high values retain their modeled status.

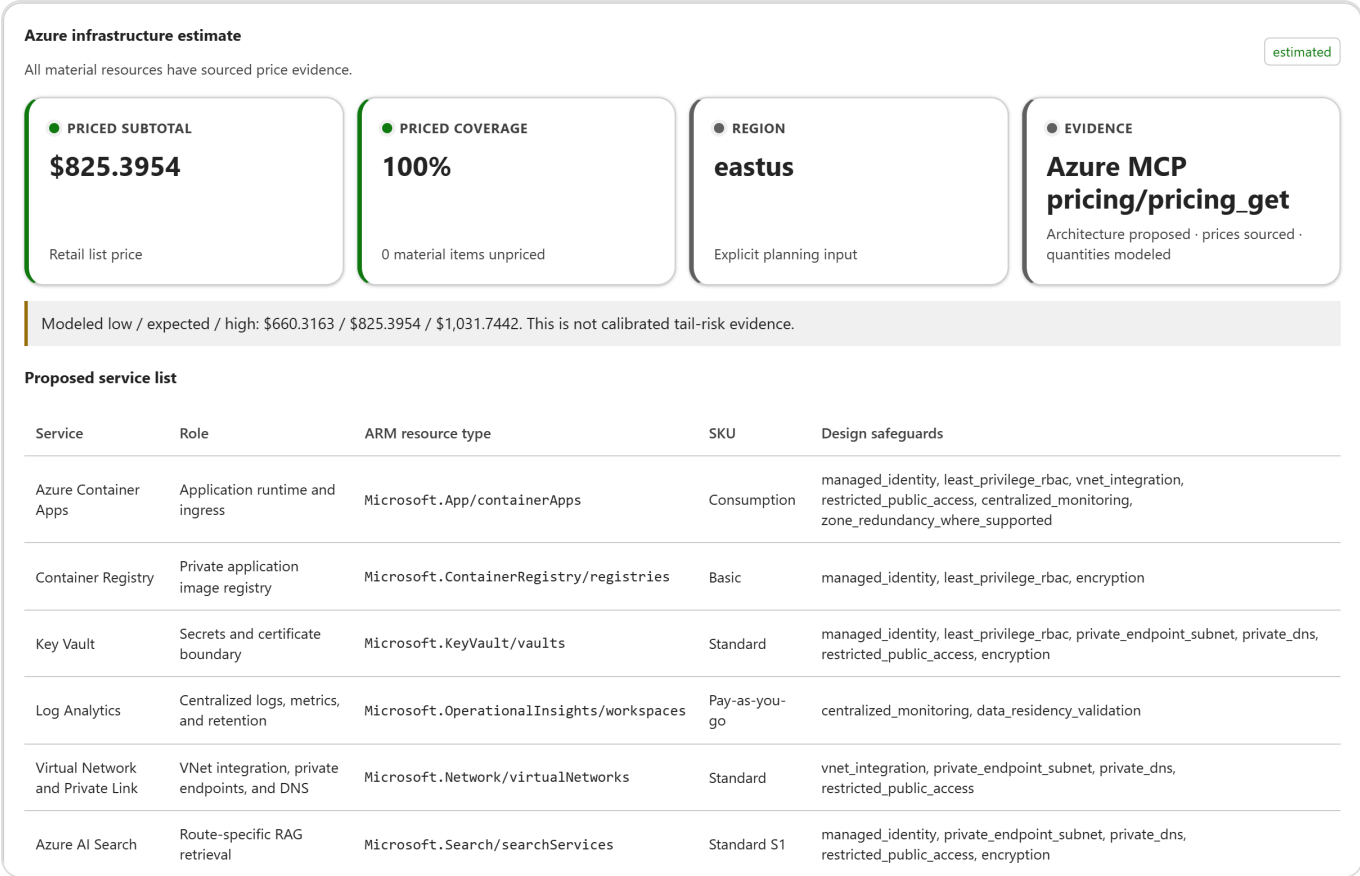
Expand the evidence panels to understand the result:

Panel	What to inspect
Token estimate and calculation evidence	Modality quantities, distribution points and the calculation explanation.
Model cost forecast evidence	Per-invocation, daily, monthly and annual ranges; Show calculation reveals the underlying assumptions.
Confirmed workload analysis	Analyzer/rule version, inferred versus confirmed topology, modalities and tools.
Non-model tool-charge evidence	Separately estimated generic tool charges; do not silently add them to measured managed-retrieval costs.
Workload, assumptions, and provenance	Prediction and receipt identity, model/catalog/pricing sources, workload and segment contracts, and method.

The stored per-call model mean is **USD 0.003424**; the period table rounds it for display. Its assumptions include five retrieved chunks of 512 tokens, a 60% system-prompt share and a 75% cache hit rate after the first call per user. These are not measurements from the ten-question batch.

Why this screen matters: it makes the forecast reproducible. Revising a misleading assumption should create new forecast evidence, not alter the old receipt to match a later observation.

5. Forecast: infrastructure is a separate estimate



Azure AI Search contributes **USD 735.84/month**: Standard S1 at USD 0.336 per search-unit hour, with three modeled replicas, one partition and 730 hours. The private-endpoint line contributes USD 73/month. The other priced lines cover the defined compute, memory, requests, registry, secrets and logging assumptions.

100% coverage is coverage of the proposed baseline, not every possible charge. It does not prove that all safeguards were deployed, that shared resources belong to this batch, or that managed retrieval and embedding charges are fully known. Retail list prices are not negotiated prices or invoices.

Why this screen matters: a very small model-call allocation can coexist with material shared infrastructure costs. Neither should hide the other, and a monthly capacity estimate must not be divided among ten questions without a defensible allocation rule.

6. Policy: authority, limits and expiry

Select **Policy**, then **Effective policy**. This page reads the configured Azure authority rather than turning the forecast's assumptions into permission.

tokengov-te003-live-proof

active

2026-09-09.measurement.1

Edit policy

Spend limits

Maximum model cost / call

\$0.02

Per-tenant run budget

\$5.00

Hard-cap action

deny

Infrastructure coverage

Policy mode

Legacy boolean

Infrastructure estimate

Required

Quality and automation

Minimum quality

0.8

Minimum segment samples

60

Consecutive breaches

2

Mutation mode

manual

Approved supply

Providers

azure_openai

Models

gpt-5.6-luna, gpt-4.1-mini

Verified pricing

Required

Optimization

Routing mode

balanced

Semantic cache

Disabled

Cache score threshold

0.83

Figure 9. The current policy is read-only. Model spend, quality settings and infrastructure requirements are separate controls.

Current control	Displayed value	How to interpret it
Policy version	2026-09-09.measurement.1	The version read during this walkthrough.
Maximum model cost per call	USD 0.02	A model admission ceiling, not a complete-trajectory cost guarantee.
Per-tenant run budget / hard-cap action	USD 5 / deny	General policy controls; do not confuse them with bounded measurement authorization.
Approved supply	azure_openai; gpt-5.6-luna and gpt-4.1-mini	Policy allowlisting does not mean either deployment is currently runnable.
Verified pricing	Required	Unsupported price assumptions cannot silently pass as verified.
Infrastructure requirement	Required; Legacy boolean mode	This particular policy has the legacy requirement, not the richer region/coverage controls available in newer policy contracts.

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Current control	Displayed value	How to interpret it
Routing / semantic cache	balanced / Disabled	Recorded controls, not proof that routing or caching optimized this batch.
Quality / segment samples	0.8 / 60	Requirements, not observed quality or samples supplied by this report.
Consecutive breaches / mutation mode	2 / manual	Response posture; no automatic optimization is implied.

Azure source	https://appcs-xbk6ickycmp22.azureconfig.io
Key / label	tokengov:policy / te003-live-v2
ETag	hXBxv0UJCUKRJRrF1r6XH0tcNy_Z9L19qCZayqy9UjA
Policy SHA-256	bdbee8e7f9c8dc578151347831ff03a8622695f3030e9f4e9c6d9c4e4c449205
Approval environment	tokengov-production
Review integration	Setup required · app_id, installation_id, repository, private_key, authenticated_ingress, authorized_principal
Publication	Approved PR merges invoke the protected workflow. Azure ETag and content are verified before the proposal is shown as Active.

Figure 10. Azure source, key/label, ETag and hash identify the authority. Hosted review integration currently reports setup required.

The ETag and content hash matter when reviewing or publishing a replacement: a proposal must not overwrite a policy that changed underneath it. Runtime readers and authorized policy publishers have separate responsibilities.

Measurement-only scope	
Authorization	Expired; cannot authorize another run
Maximum questions	10
Maximum output tokens	1024
Maximum elapsed seconds	300
Observed-model stop (USD)	0.25
Expires	2026-09-11T13:45:53.970945+00:00
Read-only configuration readiness is not execution authorization. Measurement collects incomplete-cost evidence only; a hard spend cap is not guaranteed, quality acceptance is not evaluated, and operational promotion is not permitted. Historical policy presence does not authorize new spending.	

Figure 11. The policy document is active, but its measurement grant is expired.

As of the screenshot date, another measurement run is not authorized. The grant expired on **September 11, 2026 at 13:45:53 UTC**. Its recorded bounds were 10 questions, 1,024 output tokens per response, 300 elapsed seconds and a USD 0.25 observed-model stop.

An observed-model stop can stop subsequent questions; it is **not a guaranteed all-in spend ceiling**, especially where hidden managed-service charges are unavailable. This measurement scope does not evaluate quality or permit operational promotion. Historical success is not permanent permission for future spending.

7. Policy: change requests and publication

Measurement authorization follows this same protected review flow: inspect the durable draft ID and limits, send it for approval, merge the reviewed pull request, approve publication, and verify Azure read-back. A draft, approved pull request, or read-only connection check is not permission to run. Return to Execute and check the deployed agent again after publication.

No drafts yet. Propose a policy change to begin review.

Figure 12. Change requests is an inspection surface here; no draft was created for the walkthrough.

The hosted list currently says **No drafts yet**. This is not evidence that the active policy was never published; the current authority and historical publication evidence are distinct from this hosted draft inventory.

Create policy, Edit active policy, Edit policy and **Request change** start proposal authoring. A saved draft is a durable proposed revision with a business reason, not an immediate change to Azure.

The intended protected sequence is draft, review request, reviewed pull request, approved publication and Azure read-back verification. **The hosted review integration currently reports setup required**, including its GitHub App and authenticated/authorized caller configuration. Do not interpret the explanatory publication text as proof those hosted controls are operational today.

Why this screen matters: it separates requesting a change from authorizing one. This guide neither creates a replacement grant nor suggests bypassing the expired grant. Operator setup and a separately reviewed publication would be needed before a future authorized measurement.

8. Execute & Review: understand the playground

Deployed RAG playground

Questions → deployed Foundry agent → configured hybrid retrieval. This layout comes from the saved Forecast topology, not a separate workload selection.

Check the configured books RAG before sending a question.

Measurement-only authorization must be reviewed, published and active in Azure. A draft or browser selection cannot authorize a run. Costs are partial; the observed-model stop is not a hard spend cap and total spend is not guaranteed. Quality acceptance is not evaluated and operational promotion is not permitted.

Review measurement authorization in Policy

Load 10 sample questionsDownload sample JSON

Upload questions (.txt or .json)

Choose FileNo file chosen

Questions (one per line)

Paste up to 10 questions, one per line, or upload a text file / JSON array of strings.

No questions loaded. Uploading or loading samples does not execute them.

Figure 13. The deployed RAG playground is selected by the saved forecast. The question input is deliberately left empty.

The topology badge is **rag_pipeline**. The configured path uses the deployed Foundry agent wrapper and its knowledge retrieval, not the older direct lexical Search-plus-model adapter retained in historical evidence.

Feature	Purpose	Boundary in this walkthrough
Check deployed agent	Reads configured agent/model/retrieval readiness.	Not clicked. A connection check cannot grant permission to run.
Review measurement authorization in Policy	Navigates to policy review.	Navigation, not renewal or publication.
Load 10 sample questions / download sample JSON	Helps prepare a future bounded input set.	No samples loaded; downloading or loading is not execution.
Upload questions / Questions	Accepts a text file, JSON array or one question per line.	No questions uploaded, entered or submitted.
Run measurement-only batch	Explicit billable operation, independently authorized by current Azure policy.	Not clicked; the current grant is expired.
Prepare a new batch	Resets preparation after a known outcome; does not itself run.	Not needed to inspect saved runs.
Next: Performance & Decisions	Opens the existing evidence review.	Does not evaluate quality or start optimization.

The implemented measurement input allows 1-10 questions, at most 1,200 characters per question and files up to 64 KiB. Questions run sequentially when separately authorized. Unknown submission outcomes remain locked against an accidental new attempt; saved runs should be inspected before any retry.

Why this screen matters: it makes the boundary between preparing, executing and inspecting explicit. The capability to display a playground is not evidence that all execution prerequisites are currently satisfied.

9. Execute & Review: open the completed batch

Scroll to **Runs in this report** and choose **Review batch economics** for **Run 3**, September 9, 2026 at 11:06 AM Eastern.

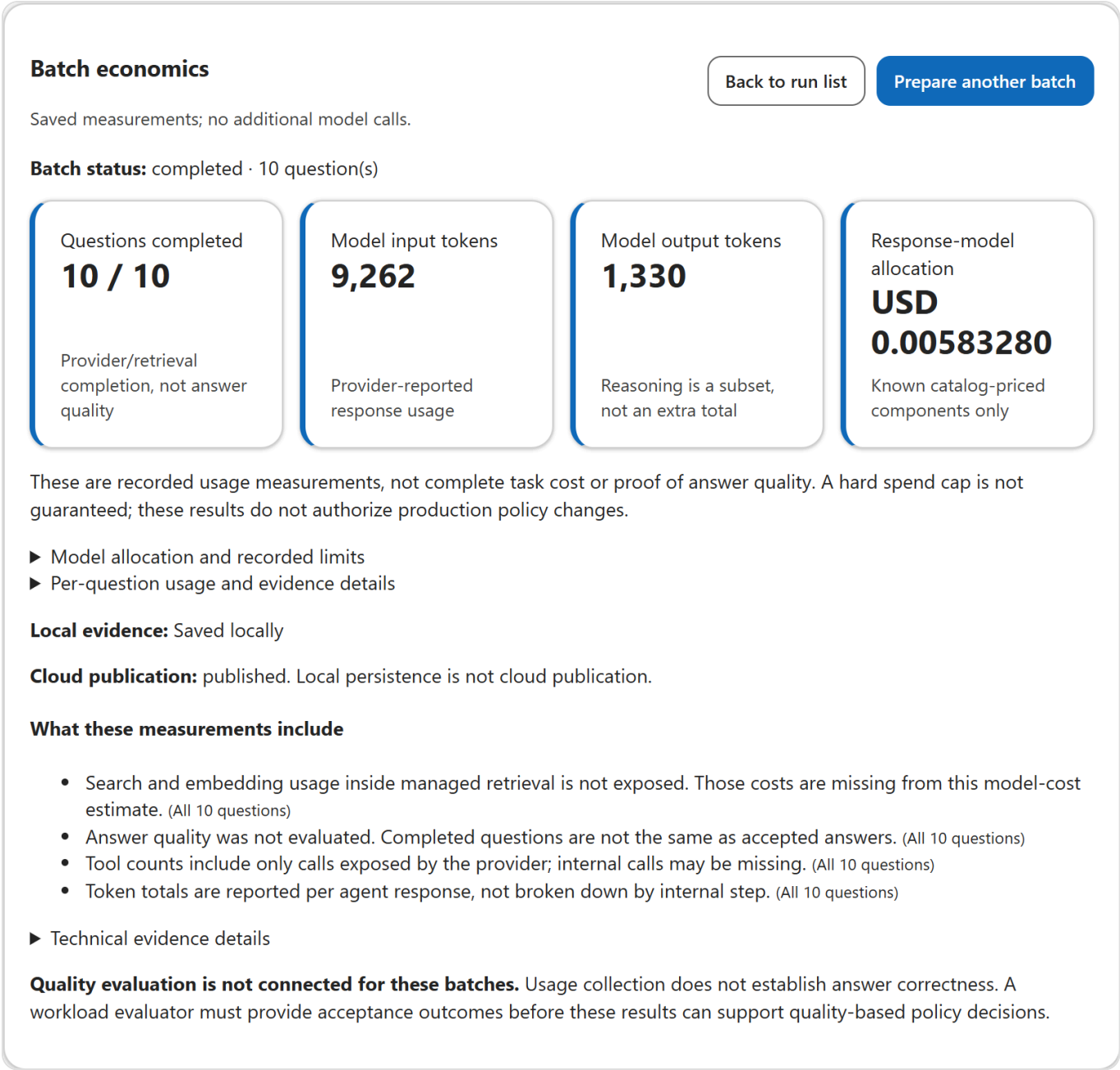
Runs in this report

Newest first. Dates use your local time; unknown dates appear last.

Run	Created / started	Status	Questions	Review
Run 3 ▶ Technical ID	Sep 9, 2026, 11:06 AM	Completed	10	Review batch economics
Run 2 ▶ Technical ID	Sep 8, 2026, 7:09 PM	Blocked	1	Review blocked attempt
Run 1 ▶ Technical ID	Sep 8, 2026, 3:23 PM	Incomplete	Not recorded	Review run

Figure 14. Run 3 is the completed ten-question example. The earlier blocked and incomplete attempts remain visible, not silently converted to successes.

The run table is newest first and exposes long technical IDs only on expansion. Different attempts are not pooled. An earlier report run can belong to a different forecast; the Performance batch selector contains only supported matches for the selected forecast.



▼ Per-question usage and evidence details										
Question #	Status	Question words / characters	Answer words / characters	Input tokens	Output tokens	Cache read / write	Reasoning tokens	Embedding tokens	Retrieval / tool calls	Latency (ms)
1	completed	10 / 66	49 / 336	927	116	0 / 0	0	Unavailable	1 / 1	11,660
2	completed	14 / 81	47 / 303	893	114	0 / 0	0	Unavailable	1 / 1	6,100
3	completed	6 / 41	41 / 274	909	91	0 / 0	0	Unavailable	1 / 1	4,430
4	completed	13 / 84	89 / 580	914	159	0 / 0	0	Unavailable	1 / 1	5,800
5	completed	9 / 50	57 / 351	910	120	0 / 0	0	Unavailable	1 / 1	5,720
6	completed	11 / 62	119 / 823	909	181	0 / 0	0	Unavailable	1 / 1	8,590
7	completed	7 / 47	41 / 280	927	91	0 / 0	0	Unavailable	1 / 1	4,850
8	completed	10 / 61	36 / 229	927	99	0 / 0	0	Unavailable	1 / 1	4,320
9	completed	13 / 83	34 / 218	1,019	89	0 / 0	0	Unavailable	1 / 1	6,570
10	completed	15 / 108	173 / 1,108	927	270	0 / 0	0	Unavailable	1 / 1	8,070

Words and characters describe the submitted question and visible answer, computed in memory; they do not measure the hidden agent prompt or retrieved context. Tokens come from provider usage at its reported scope. Cache and reasoning details are subsets, not additional totals. Missing internal-model or vectorization usage is unavailable, never inferred from words. No question text, answers, passages or tool payloads are saved in this batch's Studio evidence.

Figure 16. Per-question metrics expose variation without reconstructing retained question or answer content. Wide tables can be scrolled horizontally in Studio.

Expand **Per-question usage and evidence details** to inspect completion, word and character counts, input/output tokens, cache and reasoning subsets, available retrieval/tool counts and latency. For example, question 1 records 927 input tokens, 116 output tokens and 11,660 ms latency.

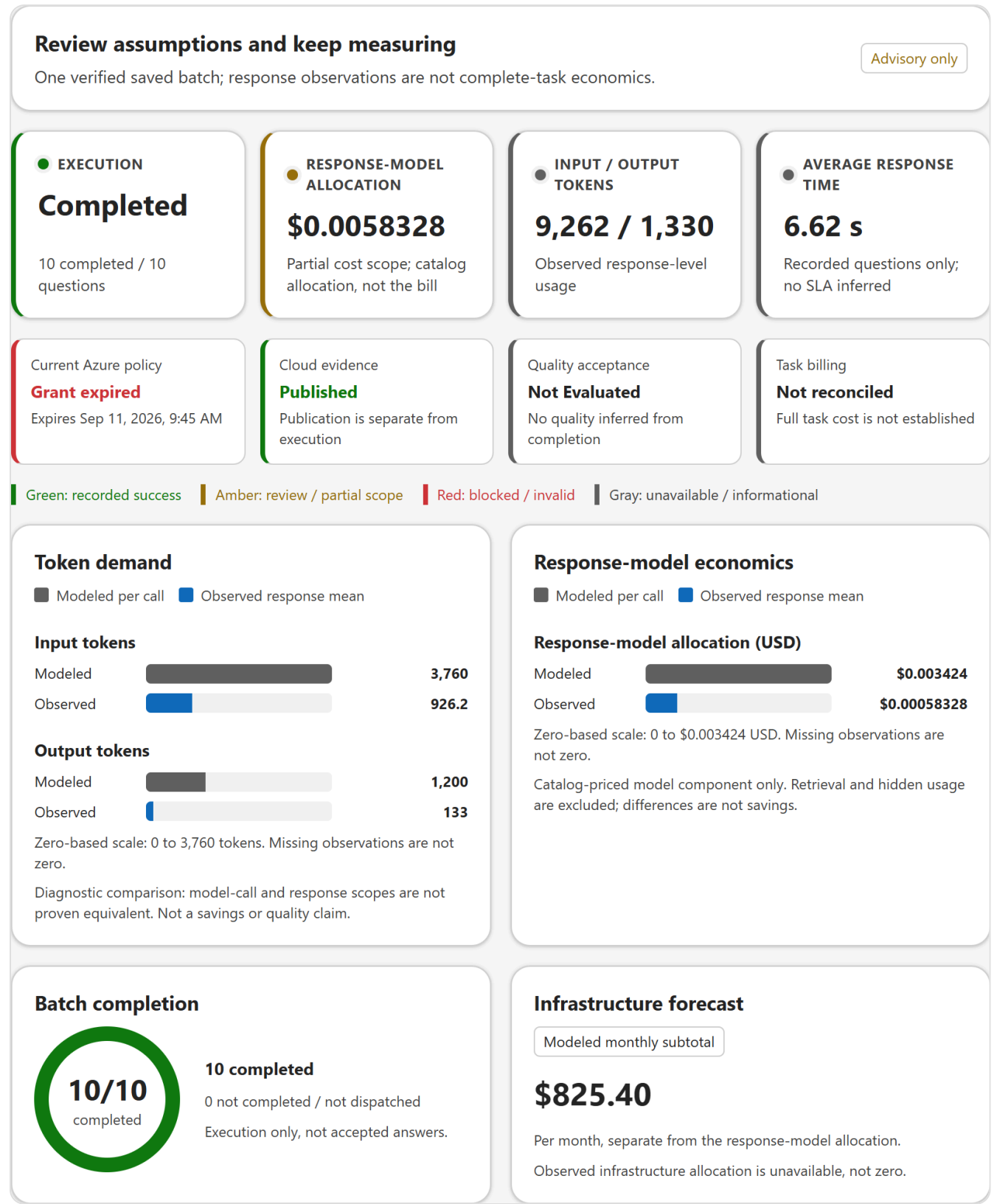
Word/character counts describe the submitted question and visible answer, computed in memory. They do not measure hidden agent prompts or retrieved context. Provider tokens are not inferred from word counts. Cache-read/write and reasoning details are subsets, not extra quantities to add to input/output totals.

Embedding usage is unavailable, not zero. Internal tool calls may be missing. The current metrics-only record does not retain question text, answers, retrieved passages or tool payload bodies. Provider-side retention and older historical records are separate boundaries.

The page states **Quality evaluation is not connected for these batches**. Compatible acceptance-evidence workflows exist for supported evidence, but metrics-only batches cannot acquire acceptance merely by opening a review.

10. Performance & Decisions: read the dashboard

Select **Performance & Decisions**. Its Actions block selects a saved batch for the exact forecast. Keep the completed September 9 ten-question batch selected. **Refresh evidence** rereads saved measurements and current policy; **Save advisory review** would create a new record and was not used.



The headline is **Review assumptions and keep measuring**, with **Advisory only**. Read the four independent status dimensions:

Status	Current display	Meaning
Execution	Completed	The selected historical batch returned ten responses.
Current Azure policy	Grant expired	New measurement is blocked despite historical completion.
Cloud evidence	Published	Evidence publication succeeded separately from execution.
Quality acceptance / task billing	Not Evaluated / Not reconciled	Neither useful-output acceptance nor full-task billing is established.

The dashboard also displays **6.62 seconds average response time**. This is the mean of recorded per-question latencies, not the elapsed batch duration or a service SLA.

The **Token demand** chart compares modeled per-call input/output with observed response means. **Response-model economics** has its own dollar scale. Both start at zero; missing observations do not get zero-height bars. **Batch completion** is a 10/10 execution ring, not a quality score. **Infrastructure forecast** is the separate USD 825.40 modeled monthly subtotal.

Colors are accompanied by labels: green for recorded success, amber for review or partial scope, red for blocked/invalid evidence, and gray for unavailable or informational evidence. A green execution card does not clear an amber or red control elsewhere.

11. Detailed expected-versus-observed comparison

Expand **Detailed expected-versus-observed comparison**, immediately below the completion/infrastructure row and before billing.

▼ Detailed expected-versus-observed comparison

Observed vs forecast: amber ↑ above, blue ↓ below, gray = same. Direction is numeric, not better/worse or a time trend. Partially comparable rows are diagnostic only.

Variance is shown only for compatible scopes. Modeled percentiles do not establish calibrated tail risk.

Measure	Forecast	Observed	Observed vs forecast	Unit / scope	Interpretation
Input tokens: modeled call / observed response mean	3,760	926.2	↓ Below Diagnostic only	tokens	Partially Comparable Diagnostic only: modeled per-call scope and cumulative response scope are not proven equivalent; no percent variance.
Output tokens: modeled call / observed response mean	1,200	133	↓ Below Diagnostic only	tokens	Partially Comparable Diagnostic only: modeled per-call scope and cumulative response scope are not proven equivalent; no percent variance.
Model cost: modeled call / observed response mean	\$0.003424	\$0.00058328	↓ Below Diagnostic only	USD	Partially Comparable Diagnostic only: modeled per-call scope and cumulative response scope are not proven equivalent; no percent variance.
Modeled tool cost per call	\$0.002500	Unavailable	Unavailable	USD/call	Not Comparable Managed retrieval/tool billing is unavailable; generic pricing applicability is unverified.
Modeled monthly infrastructure	\$825.40	Unavailable	Unavailable	USD/month	Not Comparable Monthly infrastructure is separate from per-question allocation; no observed resource allocation.
Complete task cost	Unavailable	Unavailable	Unavailable	USD/task	Unavailable Neither a complete-task forecast comparison nor complete observed cost is available.
Accepted-task quality	Unavailable	Unavailable	Unavailable	acceptance	Unavailable No evaluated segment acceptance outcomes; completed-question counts are not quality.

Figure 18. Direction is visible, while the scope column prevents a misleading savings or accuracy claim.

Comparable-looking component	Forecast	Observed response mean	Reading
Input tokens	3,760	926.2	Below; diagnostic only.
Output tokens	1,200	133	Below; diagnostic only.
Response-model allocation	USD 0.003424	USD 0.00058328	Below; diagnostic only.

The blue down-arrow means the observed number is below the forecast number. An amber up-arrow means above, and gray equals means the same. These are numeric directions using unrounded values, **not time trends or better/worse judgments**. The actual screenshot shows the below cases; it is not fabricated to demonstrate all three directions.

The table labels these pairs **Partially Comparable** and withholds percentage variance because modeled call scope and provider-response scope are not proven equivalent. The observed means divide the saved totals by ten completed responses; they do not represent complete agent-trajectory costs.

Tool and infrastructure observations are unavailable. Complete task cost and accepted-task quality are unavailable on both sides. None should become zero or be omitted to make a total appear complete.

Why this screen matters: it reveals assumptions worth reviewing without claiming that a smaller answer preserved quality. In this example, the 1,200-token forecast output assumption also exceeds the recorded 1,024-token response cap; that deserves review, not a savings headline.

12. Billing reconciliation

Expand **Billing reconciliation and forecast learning**, immediately after the detailed comparison. This section is bound to the selected forecast and batch, not all historical experiments.

Billing imported; allocation incomplete

Partial. Full task billing is not established.

SOURCE BILLING ROWS

216

Bound resource scope, not this batch's bill

SOURCE-PERIOD COST

USD 33.671969

Resource charges; currencies stay separate

Batch date: 2026-09-09 (UTC). Source period: 2026-09-01 to 2026-09-09.

Last import: Sep 9, 2026, 3:51 PM. Source: Azure Cost Management Export.

Export delivered: Sep 9, 2026, 6:45 AM. **This export predates execution; it cannot contain this batch's complete charges.**

Batch-day resource total: USD 1.696462. Unallocated: USD 1.696462.

Shared infrastructure, retrieval and model charges are not assigned to this batch without a measured allocation rule. Current-period Azure costs may change. Catalog response-model allocation is separate, not added again.

Resource role	Allocation	Batch-day resource costs
Studio Hosting	Unallocated	USD 0.06282
Policy Configuration	Unallocated	USD 1.2
Model And Embedding	Missing Billing Data	Unavailable
Image Registry	Unallocated	USD 0.027767
Observability	Unallocated	USD 0.001864
Managed Retrieval	Unallocated	USD 0.404
Studio Persistence	Unallocated	USD 0.000003
Evidence And Exports	Unallocated	USD 0.000008

Figure 19. Actual imported resource charges are visible, but no allocation to the ten-question batch is invented.

Billing evidence	Recorded value	Meaning
Bound source rows	216	Rows for configured resource scope; 887 other rows were excluded from the 1,103-row source export.

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Billing evidence	Recorded value	Meaning
Source period	September 1-9, 2026	Period represented by the imported snapshot, not the batch duration.
Source-period charges	USD 33.671969	Rounded display of USD 33.67196892192 for bound resources.
September 9 resource charges	USD 1.696462	Rounded display of USD 1.69646161512 for that resource day.
Unallocated September 9 charges	USD 1.696462	The entire known resource-day amount remains unallocated to this batch.
Model and embedding resource	Missing Billing Data	Required billing evidence is missing; this is not a zero-cost line.

The export was delivered at **10:45:38 UTC on September 9**, before the batch started at **15:06 UTC**. It cannot establish this batch's complete charges. Neither USD 33.67 nor USD 1.70 is the cost of the ten questions. The USD 0.0058328 catalog model allocation is a separate view and must not be added to potentially overlapping billing rows.

Cost Management is integrated, but current hosted source access is not fully operational. A real snapshot has been imported and remains readable. As recorded in the deployment decisions, hosted resync is blocked by the billing source's network restrictions; having Blob Data Reader alone does not bypass networking. This guide did not attempt a new sync.

Explicit action	Intended effect	Does opening the panel do this?
Record usage feedback	Registers verified response usage and saves partial feedback using already imported billing.	No.
Sync billing & reconcile	Reads the configured Azure export, imports a source revision and records partial reconciliation/feedback. Requires source access and authorization.	No.

Both actions write evidence even though neither runs questions. Replaying unchanged input is deduplicated, not counted as new training evidence. Shared resource allocation remains server-controlled and requires a measured allocation rule.

13. Response feedback, learning and WAPE

The lower card in the same billing/learning panel explains whether saved response measurements are eligible to improve future predictions.

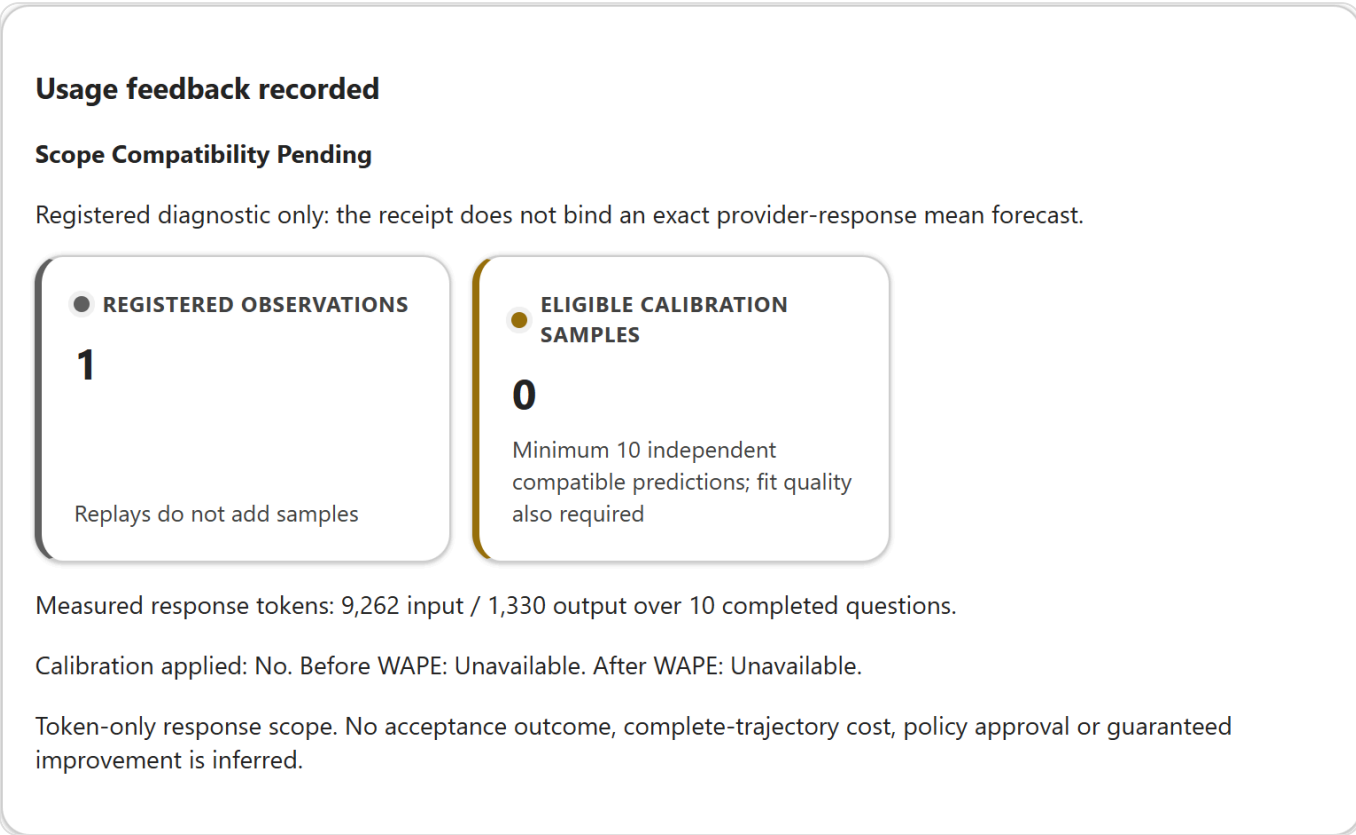


Figure 20. Recorded observations and eligible learning samples are deliberately different counts.

The current card shows **Usage feedback recorded**, **Scope Compatibility Pending**, **1 registered observation**, **0 eligible calibration samples**, and **Calibration applied: No**. Before and after WAPE are **Unavailable**.

The original forecast does not bind an exact provider-response mean prediction contract. Recording response usage cannot retrospectively change its scope. Ten questions under this one prediction are not ten independent matched forecast observations.

A future response-calibration path needs a compatible forecast-time `response-prediction-contract.v1`, independent observations, at least ten eligible predictions and sufficient fit quality. A separate held-out evaluation is needed to demonstrate improvement. These requirements are not satisfied by this batch, and response calibration would still not establish complete-task quality or cost.

What WAPE means

Select the **i** beside **Forecast error (WAPE)** to open the help note.

What is WAPE?

Weighted Absolute Percentage Error measures how far forecasts differ from actual usage.

$$\text{WAPE} = \text{sum of absolute forecast errors} / \text{sum of actual values} \times 100.$$

Lower is better; 0% means an exact match. It can exceed 100%: forecasting 220 tokens when actual usage is 100 tokens gives 120% WAPE. It does not tell you whether forecasts were too high or too low.

These learning records compare **token usage**, not dollar cost or answer quality. Before and after compare the original and subsequent forecasts with the same measured samples. An unchanged result means no demonstrated improvement; it does not prove the forecast is accurate.

Missing evidence is unavailable, not zero. WAPE is undefined when total actual usage is zero. This metric is not a calibrated budget-risk guarantee.

Close

Figure 21. WAPE help is accessible without running or registering anything.

WAPE = sum of absolute forecast errors / sum of actual values x 100.

For example, forecasts of 120 and 220 tokens versus actuals of 100 and 200 tokens give $(20 + 20) / (100 + 200) \times 100 = 13.33\%$. The denominator gives larger actual volumes more weight; this is not the simple average of the individual percentage errors.

Lower is better; 0% is an exact match on the compared samples. Forecasting 220 tokens against 100 actual tokens gives **120% WAPE**, which is valid. WAPE does not show whether errors were too high or too low. It is undefined when total actual usage is zero.

Here WAPE is **token error**, not billing accuracy, answer quality, completion percentage or calibrated budget risk. **Unavailable** is not 0%. Equal before and after values mean no demonstrated improvement, not proof of accuracy. Close the help with **Close**, Escape or an outside click.

14. Findings, policy rationale and saved reviews

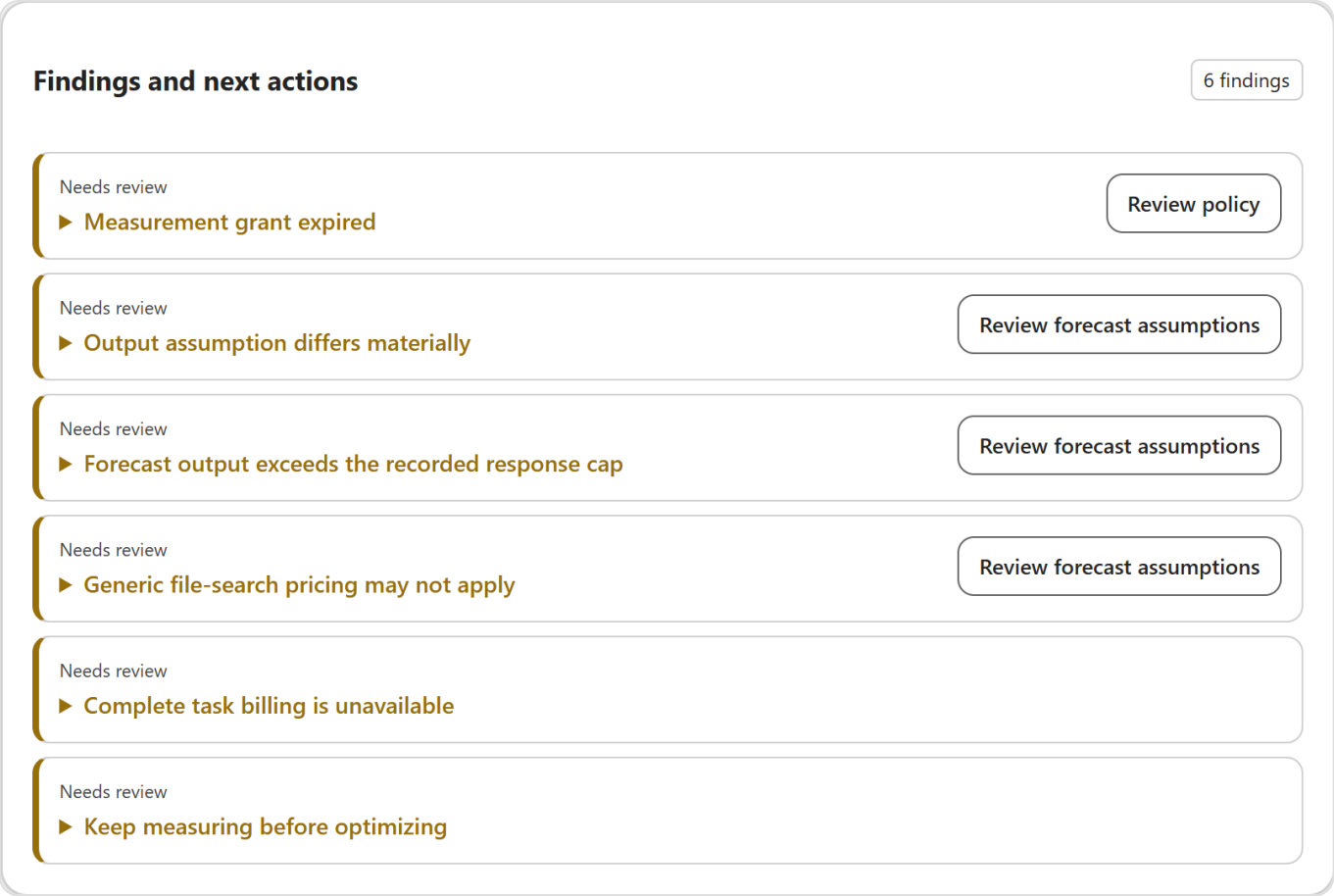


Figure 22. Findings identify what deserves attention without providing misleading one-click fixes for missing integrations.

The live review has six findings:

Finding	Meaning and appropriate next work
Measurement grant expired	Review Policy before a future authorized measurement. A historical run is not invalidated by today's expiry.
Output assumption differs materially	Review forecast assumptions against response observations, while retaining the scope caveat.
Forecast output exceeds the recorded response cap	Align future assumptions and authorization bounds through reviewed revisions.
Generic file-search pricing may not apply	Establish the managed knowledge-base retrieval meter rather than reusing an unrelated tool rate.
Complete task billing is unavailable	Obtain missing cost coverage and allocation evidence; there is no fabricated total or placeholder fix button.
Keep measuring before optimizing	Provide representative segment-level quality/acceptance evidence before making quality-preserving optimization claims.

Expand a finding to read its explanation. Existing **Review policy** and **Review forecast assumptions** links lead to the relevant surface; they were not used to change anything for this guide. Missing integrations remain explicit gaps.

Policy binding and advisory decision rationale reports the execution and current versions as the same: 2026-09-09.measurement.1. That equality does not prevent a time-bound grant from expiring. Version equality and current validity are different checks.

▼ Saved advisory reviews (1)

▼ Sep 9, 2026, 1:26 PM · Review assumptions and keep measuring

This evidence supports advisory review only. Obtain segment quality/acceptance and full cost evidence before considering optimization; future execution requires current authorization.

Historical snapshot, not current authorization. Recorded by local-operator.

- **Measurement grant nearing expiry:** The recorded and current grant must be reviewed before future measurement. This does not invalidate historical observations.
- **Output assumption differs materially:** Modeled output per call differs from the observed response average. This is an assumption diagnostic, not savings or evidence of equivalent quality.
- **Forecast output exceeds the recorded response cap:** The modeled mean output is above the measurement max-output limit. Review the forecast scope and bound; do not infer achieved quality.
- **Generic file-search pricing may not apply:** The receipt models generic file search. Managed knowledge-base retrieval has a different unverified billing scope; the receipt's complete-tool-cost flag does not prove full tool cost.
- **Complete task billing is unavailable:** Retrieval, embedding, hidden calls, evaluation and infrastructure allocation are not fully attributed. Known response-model allocation is catalog-priced, not an invoice.
- **Keep measuring before optimizing:** Quality and segment-level acceptance are not evaluated here. Completed questions do not establish acceptance, quality preservation or permission to optimize.

Figure 23. A saved September 9 review retains its original findings; it is not rewritten to reflect September 14.

There is **one saved advisory review**, recorded by **local-operator**. Its old wording says the grant was *nearing expiry*, while today's live finding says *expired*. That is expected: immutable historical evidence and a current assessment answer different questions.

Saving a new review would append an actor-attributed snapshot, not publish policy, run questions or train the predictor. **Saved quality reviews**, when applicable to compatible acceptance evidence, are separate from advisory reviews. They are not decision-grade evidence available for this metrics-only batch.

15. Project history and optional Work IQ evidence

At the bottom of **Performance & Decisions**, expand **Project history: reconciliation, learning and other workloads**.

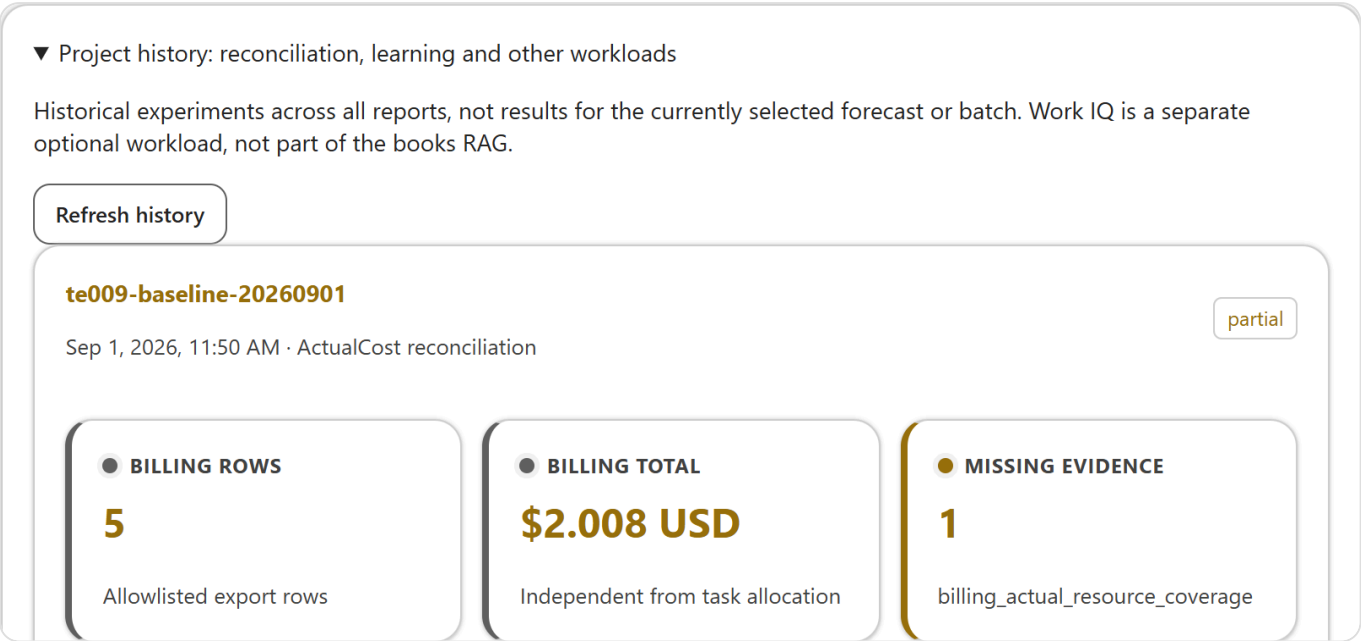


Figure 24. Historical experiment evidence remains accessible without occupying Home or being mistaken for the current batch. The screenshot shows the beginning of the expanded history.

The current history includes an older September 1 reconciliation, an unchanged learning proof and a separate **m365-calendar-count** portability experiment. Those records do not belong to the selected September 9 ten-question batch.

In particular, the older learning proof shows **121.6% before and after WAPE** on 120 measured tasks. That is not the current batch's WAPE; the current batch has no eligible calibration samples and unavailable WAPE.

Work IQ is optional and not part of this books RAG. The historical calendar-count experiment demonstrates a different workload using reusable contracts. Its one accepted task and independent ground-truth comparison do not establish statistical sufficiency or known Copilot Credit consumption for RAG.

Why this screen matters: history supports auditability and portability without letting unrelated success, acceptance rates or billing totals contaminate the current report's conclusions.

16. Safe walkthrough and current limitations

For a demonstration using saved evidence only:

- 1. Open RPT-20260908-CA037FE9 from Home.
- 2. Keep Forecast 3, prediction 5176, selected; inspect profile, economics and infrastructure.
- 3. Read Effective policy, its measurement expiry and Change requests without creating a draft.
- 4. Open Execute & Review and review Run 3; do not prepare or run questions.
- 5. Open Performance & Decisions with the same completed batch selected.
- 6. Expand comparison, billing/learning, WAPE help, findings, saved review and Project history.

Do not use Create report, Save report, Create revised forecast, forecast confirmation, policy-authoring/publication, Run, Record usage feedback, Sync billing & reconcile, or Save advisory review as part of a read-only tour. Some do not spend model tokens, but they still change evidence or authority.

Current state	Consequence
Hosted Foundry-only release	Other delivery routes are visible future-release options, not available hosted integrations.
Expired measurement grant	No new batch is authorized by that historical grant.
Hosted review integration needs setup	The protected proposal-to-publication path requires operator configuration; page visibility is insufficient.
Billing source networking remains restricted	Existing imported evidence is readable; successful fresh hosted sync is not demonstrated.
No segment acceptance outcomes for this batch	Completed responses cannot support a quality-preserving optimization claim.
Partial cost and no exact response prediction contract	No complete-task bill, accepted-task unit economics or calibrated improvement is established.

The hosted service has separate process liveness and persistent-storage readiness. A healthy application does not imply an active execution grant, accessible billing source or accepted-task evidence. If report data becomes unavailable, do not recreate reports, clear evidence or change Azure security settings to make a demonstration look complete; the operator should investigate the dependency. Deployment decision D95 records an outstanding governance-owner dependency for durable persistence availability.

Alignment with the full TokenEconomics lifecycle

This guide documents the existing **predict -> compare policy -> admit -> execute -> evaluate -> respond -> reconcile -> learn** responsibilities; it does not claim that every link is proven by this one batch. It consumes versioned saved receipts, policy provenance, sealed response measurements, billing snapshots and feedback. Its outputs are documentation and screenshots, not new runtime or decision evidence.

The current example demonstrates saved forecasting, Azure authority inspection, historically authorized measurement, response-economics review, advisory findings and partial reconciliation/feedback. Complete-trajectory costs, segment-specific acceptance, representative tail-risk evidence, authorized regression/reversion proof and improved held-out future predictions remain gaps against the [constitution's completion definition](#). No savings, quality preservation or production-validated optimization is guaranteed.

17. Evidence references and terminology

Exact example binding

Evidence	Identifier
Report	RPT-20260908-CA037FE9
Latest forecast	e997e27b-ac11-49c6-bbf9-bdc8eb3c9bd6
Prediction	5176
Original receipt	p1an_3c01293376656dbf20cd
Completed batch	run-f87aa8eae8ef4c57a05a6defb5da9b79
Batch contract	rag-agent-batch.v2
Recorded/current policy version at capture	2026-09-09.measurement.1

Modeled means calculated under stated assumptions. **Measured** means recorded usage at its documented scope. **Proposed** means not yet authoritative. **Partial** means known evidence with explicit gaps. **Unavailable** is not zero. **Advisory** is not admission, accepted-task evaluation or policy publication. **Production-validated** is not the classification of this reference walkthrough.

Source documents and implementation

- [TokenEconomics Constitution](#), especially lifecycle, authority, immutable evidence and aligned completion.
- [Decision log](#): D87-D91 for the current workflow; D92 for billing/response learning; D93-D94 for deployment and hosted route scope; D95 for current readiness and remaining operational dependencies.
- [Studio UI](#): screen labels, controls, rendering and WAPE help.
- [Studio API](#): read/write boundaries and hosted delivery enforcement.
- [Performance evidence](#) and [advisory review](#): selected-batch interpretation.
- [Batch feedback](#), [billing ingestion](#) and [response learning](#): source joins, allocation and eligibility.
- [Microsoft Copilot Credits research](#): separate meter families and source references; not a claim of hosted route availability.
- [Microsoft Learn: create and manage Cost Management exports](#): billing exports as a source, separate from workload allocation.

Screenshots use the Standard appearance and are stored in `docs/images/token-studio-guide/2026-09-14/`. Displayed values may be rounded. Later policy, pricing, billing or report revisions can change current views; historical receipts and sealed results retain their original meaning.